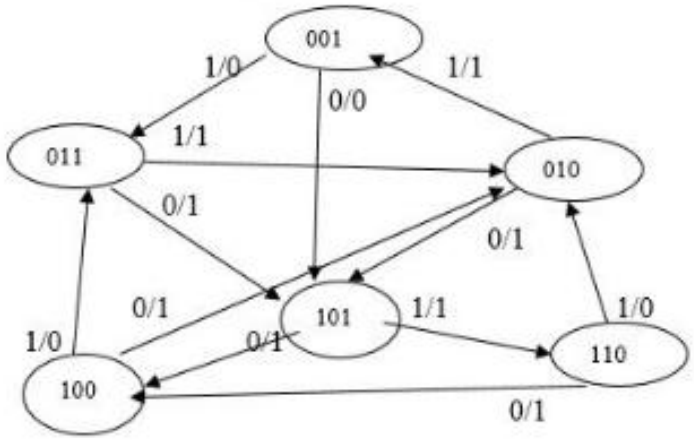
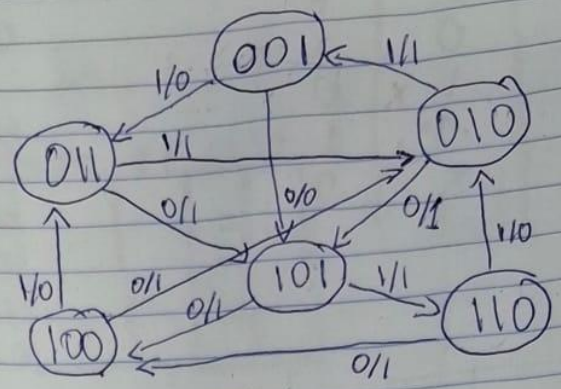
Jesús Osiris Santana Taveras – A00111974
Circuito secuencial

Universidad APEC
UNAPEC
Lógica Digital I (TEC190)
Segundo Parcial 54

Nombre:_____Fecha:_____
Profesor: Ing. Edward A. Báez G. Grupo: 51051

1) Diseñe y simule un circuito lógico secuencial que cumpla con el siguiente diagrama de estado utilizando flip flops JK.





EA

A	B	C	X
0	0	1	0
0	0	1	1
0	1	0	0
0	1	0	1
0	1	1	0
0	1	1	1
1	0	0	0
1	0	0	1
1	0	1	0
1	0	1	1
1	1	0	0
1	1	0	1

ES

A	B	C	Y
1	0	1	0
0	1	1	0
1	0	1	1
0	0	1	1
1	0	1	1
0	1	0	1
0	1	0	1
0	1	1	0
1	0	0	1
1	1	0	1
1	0	0	1
0	1	0	0

Q	Q^+	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Tabla excitación JK (Ejemplo)

J_A	K_A	J_B	K_B	J_C	K_C
1	X	0	X	X	0
0	X	1	X	X	0
1	X	X	1	1	X
0	X	X	1	1	X
1	X	X	1	X	0
0	X	X	0	X	1
X	1	1	X	0	X
X	1	1	X	1	X
X	0	0	X	X	1
X	0	1	X	X	1
X	0	X	1	0	X
X	1	X	0	0	X

JA

si hubiera hecho grupo 4x4 = x' directamente

AB \ Cx	00	01	11	10
00	X	X	0	1
01	1	0	0	1
11	X	X	X	X
10	X	X	X	X

$J_A = C'x' + Cx'$ $\quad (C' + C) = 1$

$= x'(C' + C)$

$J_A = x'$

K_A

AB \ Cx	00	01	11	10
00	X	X	X	X
01	X	X	X	X
11	0	1	X	X
10	1	1	0	0

$K_A = C'x + C'AB'$

$= \bar{C}(x + A\bar{B})$

J_B

AB \ Cx	00	01	11	10
00	X	X	1	0
01	X	X	X	X
11	X	X	X	X
10	1	1	1	0

$J_B = C' + Cx$

$= C' + x$

24

K_B

AB \ Cx	00	01	11	10
00	X	X	X	X
01	1	1	0	1
11	1	0	X	X
10	X	X	X	X

$K_B = x' + A'C'$

J_C

AB \ Cx	00	01	11	10
00	X	X	X	X
01	1	1	X	X
11	0	0	X	X
10	0	1	X	X

$A' + AB'X$

$J_C = \bar{A} + \bar{B}X$

K_C

AB \ Cx	00	01	11	10
00	X	X	0	0
01	X	X	1	0
11	X	X	X	X
10	X	X	1	1

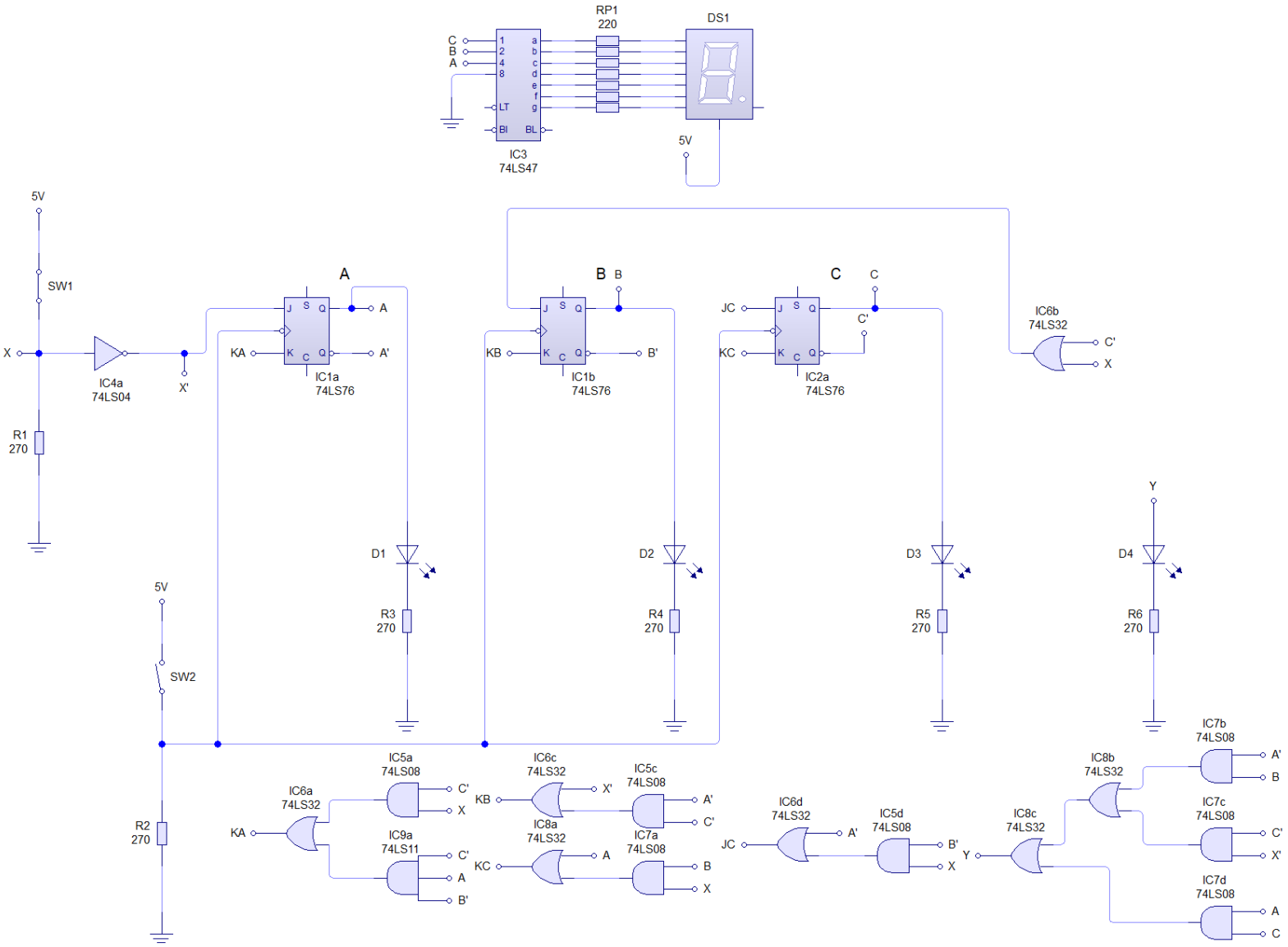
$A + A'BX$

$K_C = A + BX$

y

AB \ Cx	00	01	11	10
00	X	X	0	0
01	1	1	1	1
11	1	0	X	X
10	1	0	1	1

$A'B + C'x' + AC$


RP1
220
DS1
C
B
A
1
2
4
8
a
b
c
d
e
f
g
LT
BI
BL
IC3
74LS47
5V
5V
SW1
X
IC4a
74LS04
X'
R1
270
A
B
C
J
S
Q
K
C
Q
KA
A'
IC1a
74LS76
KB
B'
IC1b
74LS76
JC
KC
C'
IC2a
74LS76
IC6b
74LS32
C'
X
D1
D2
D3
D4
Y
R3
270
R4
270
R5
270
R6
270
5V
SW2
R2
270
IC6a
74LS32
KA
IC5a
74LS08
C'
X
IC9a
74LS11
C'
A
B'
IC6c
74LS32
KB
X'
IC8a
74LS32
KC
A
IC5c
74LS08
A'
C'
IC7a
74LS08
B
X
IC6d
74LS32
JC
A'
IC5d
74LS08
B'
X
Y
IC8c
74LS32
IC8b
74LS32
IC7b
74LS08
A'
B
IC7c
74LS08
C'
X'
IC7d
74LS08
A
C